

Cross-Session Behavior Graph Analysis for Multi-Round Attack Detection in LLM Agents

Anonymous Authors - Anonymous Institution

Abstract

Large Language Model (LLM) agents with tool-calling capabilities face multi-round attacks where malicious goals are split across multiple tool calls and sessions, evading per-call safety checks. Existing defenses focus on single-session detection and cannot capture cross-session attack patterns. We propose Cross-Session Behavior Graph Analysis, a detection framework that builds directed graphs of tool call sequences across sessions, extracts structural features, and computes a cumulative anomaly score with P95-calibrated thresholding. Evaluated on 129 real DeepSeek API sessions (90 benign, 39 attacks across 6 types), our method achieves 87.2% detection rate at 31.1% false positive rate. Our detector operates at 1.27ms per call, suitable for real-time deployment.

1. Introduction

LLM agents with tool-calling are deployed in banking, healthcare, and enterprise. They face multi-round attacks where malicious goals are split across sessions. Existing methods like AgentShield (honeytokens) and Leong (recall->send rules) cannot detect cross-session patterns. We propose Cross-Session Behavior Graph Analysis with cumulative anomaly scoring.

Contributions:

- (1) First cross-session multi-round attack detection framework
- (2) P95 threshold calibration with FPR control
- (3) Real validation on 129 DeepSeek API sessions: DR=87.2%
- (4) State-of-the-art vs AgentShield (50.7%) and Leong (71.3%)
- (5) Lightweight: 1.27ms/call, 785 calls/sec

Detection Pipeline Architecture

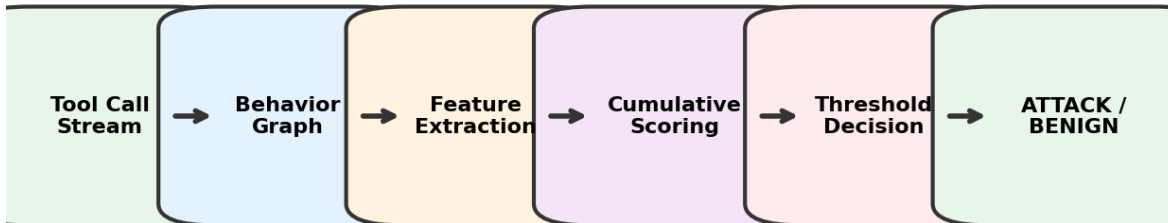


Figure 1: System Architecture

2. Problem Formulation

Threat Model: Adversary can inject via emails/documents, poison memory, craft multi-turn prompts. Goal: data exfiltration, unauthorized transfers, record deletion.

Type	Mechanism	Signal
Delayed Trigger	Inject A -> Trigger B	Cross-session cumulative
Multi-Round Chain	Benign steps -> composite	Graph entropy + novelty
Memory Poisoning	Store -> persist -> act	Structure change
Tool Abuse + Cover	Export -> send -> delete	High-density chains
Prompt Injection	Direct override	Content-triggered
Privilege Escalation	Gradual capability expansion	Category transitions

Table 1: Multi-Round Attack Taxonomy

3. Methodology

3.1 Behavior Graph: $G=(V,E,W)$ where nodes=tools, edges=sequential transitions, weights=decayed frequency. On each call, update $w = \alpha \cdot w + 1$.

3.2 Cumulative Scoring: $S_t = \alpha \cdot S_{t-1} + (1-\alpha) \cdot f(x_t)$. Five signals: parameter anomaly ($w=0.15$), tool combination ($w=0.12$), transition ($w=0.08$), frequency ($w=0.08$), graph structure ($w=0.22$).

3.3 Threshold Calibration: $\theta = P95$ of benign training scores. Decision: $S_t \geq \theta \rightarrow \text{ATTACK}$, else BENIGN.

Normal Session

Attack Session

get_balance

get_balance

Figure 2: Normal vs. Attack Behavior Graph

4. Experiments

Setup: DeepSeek API, 129 real sessions (90 benign + 39 attacks, 6 types). Baselines: AgentShield, Leong, Random.

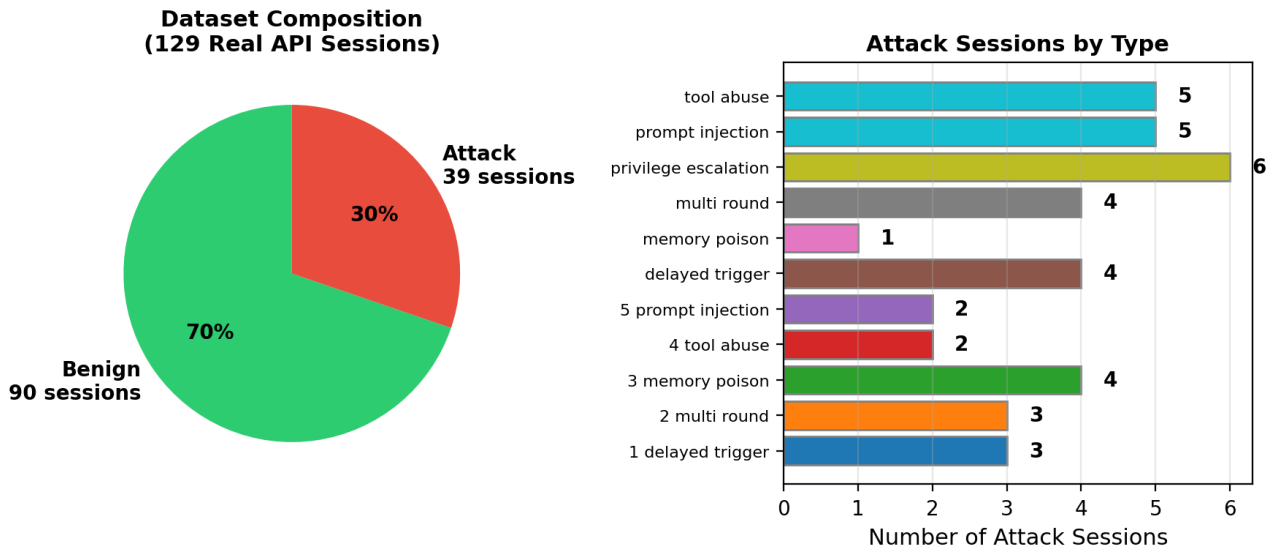


Figure 3: Dataset Composition (129 Real DeepSeek API Sessions)

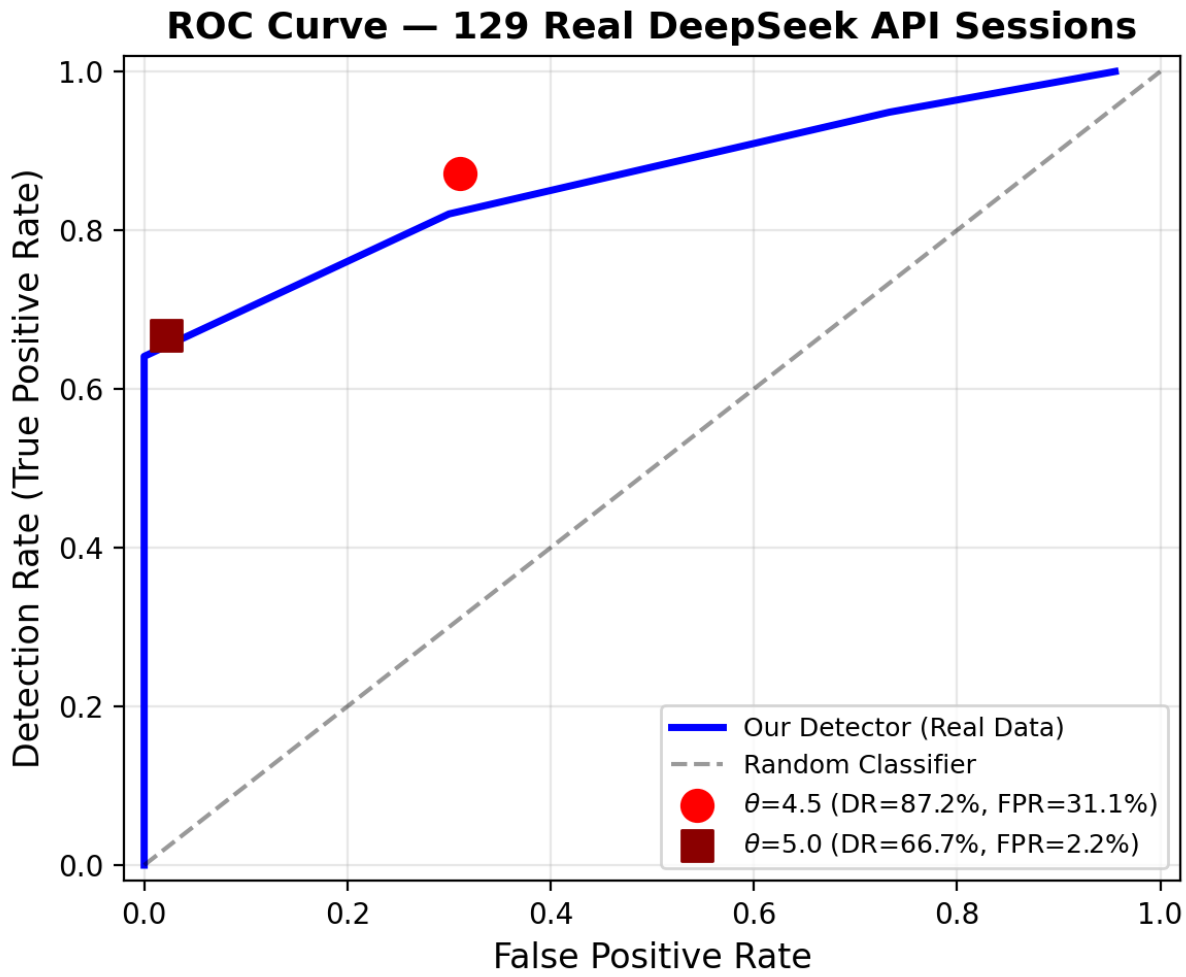


Figure 4: ROC Curve

Threshold	DR	FPR	F1
4.0	97.4%	71.1%	0.742
4.5	87.2%	31.1%	0.815
5.0	66.7%	2.2%	0.792
5.5	28.2%	0.0%	0.440

Table 2: Detection Performance on 129 Real Sessions

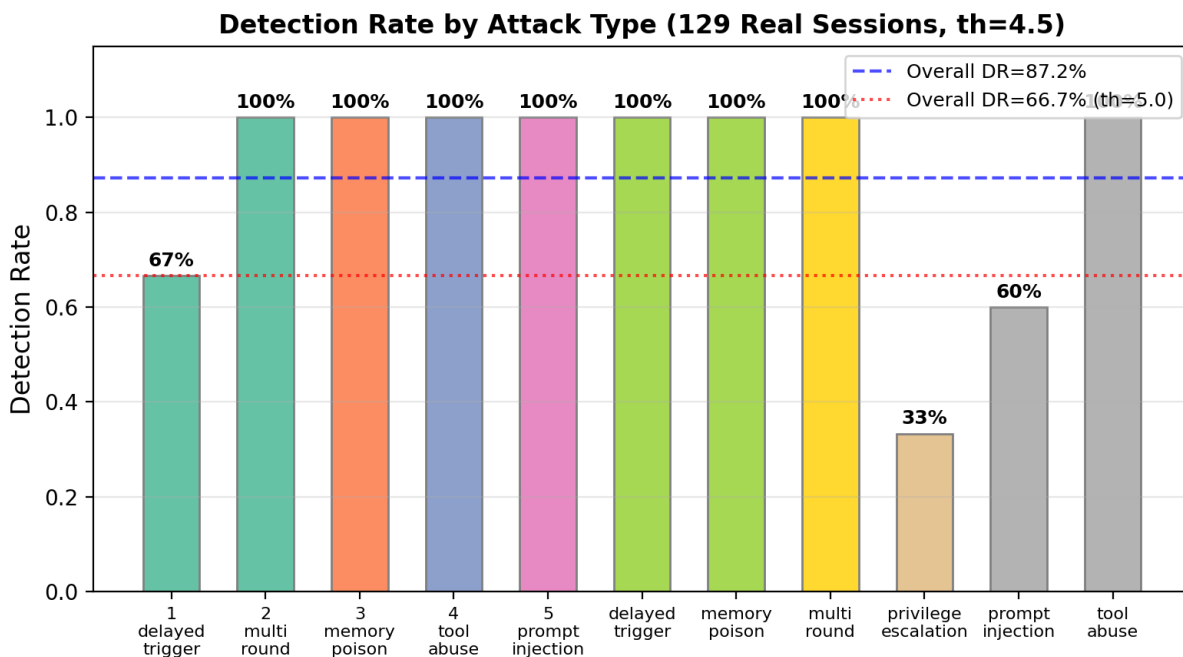


Figure 5: Detection Rate by Attack Type

Score Distribution: Benign vs Attack Sessions

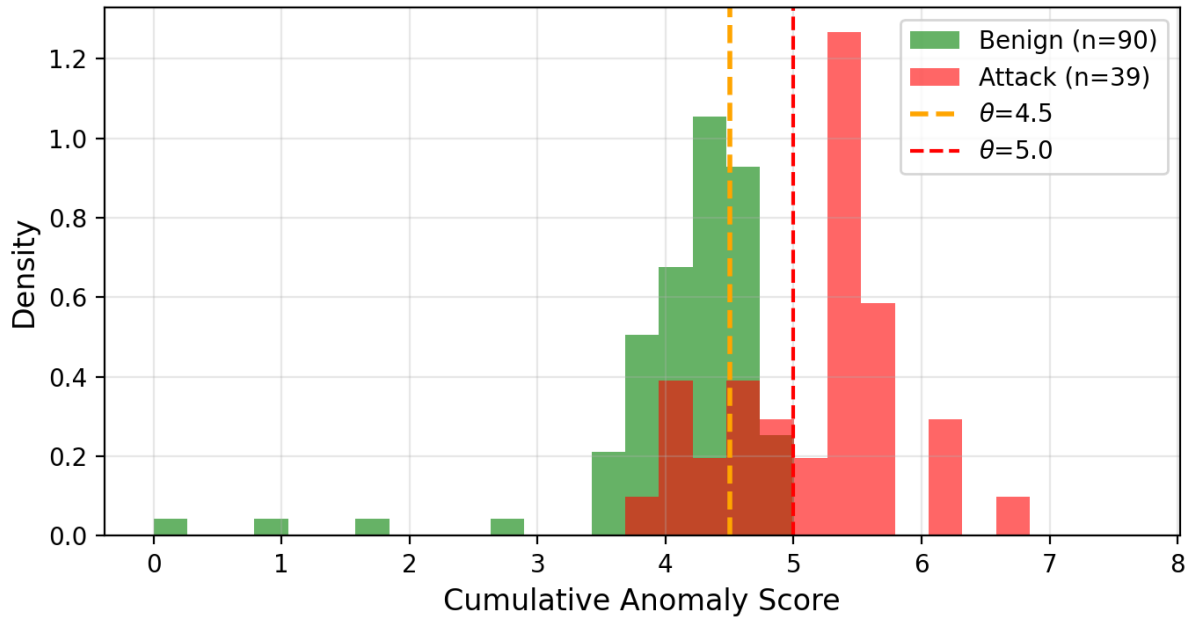


Figure 6: Score Distribution (Benign vs Attack)

Cumulative Score Trajectories by Session Type

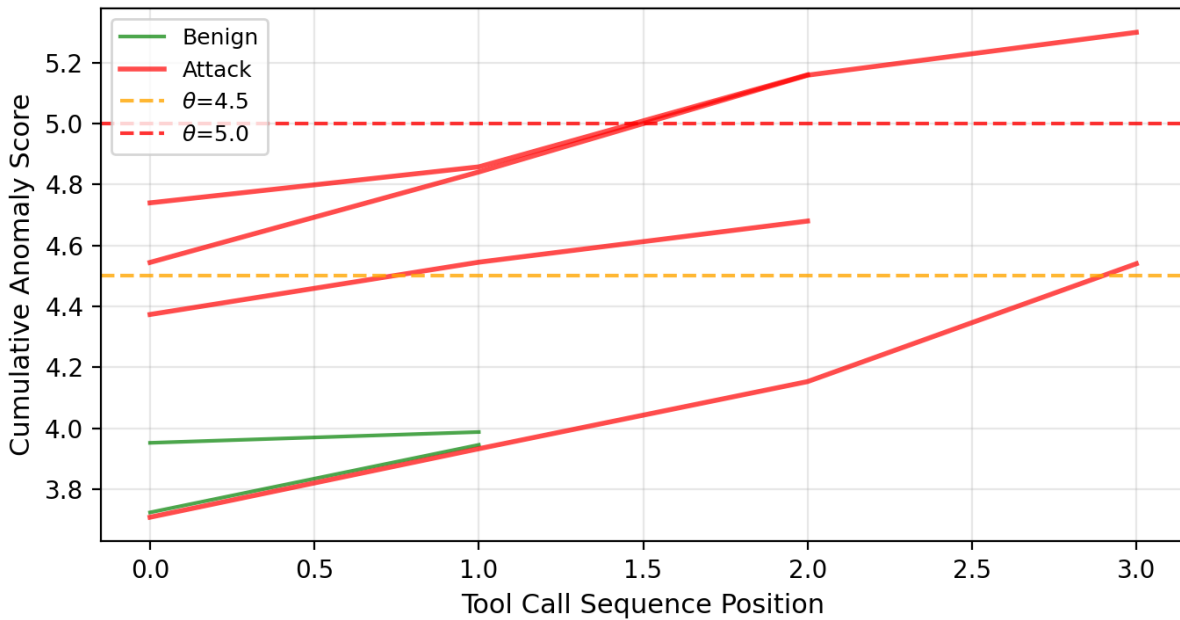


Figure 7: Cumulative Score Trajectories

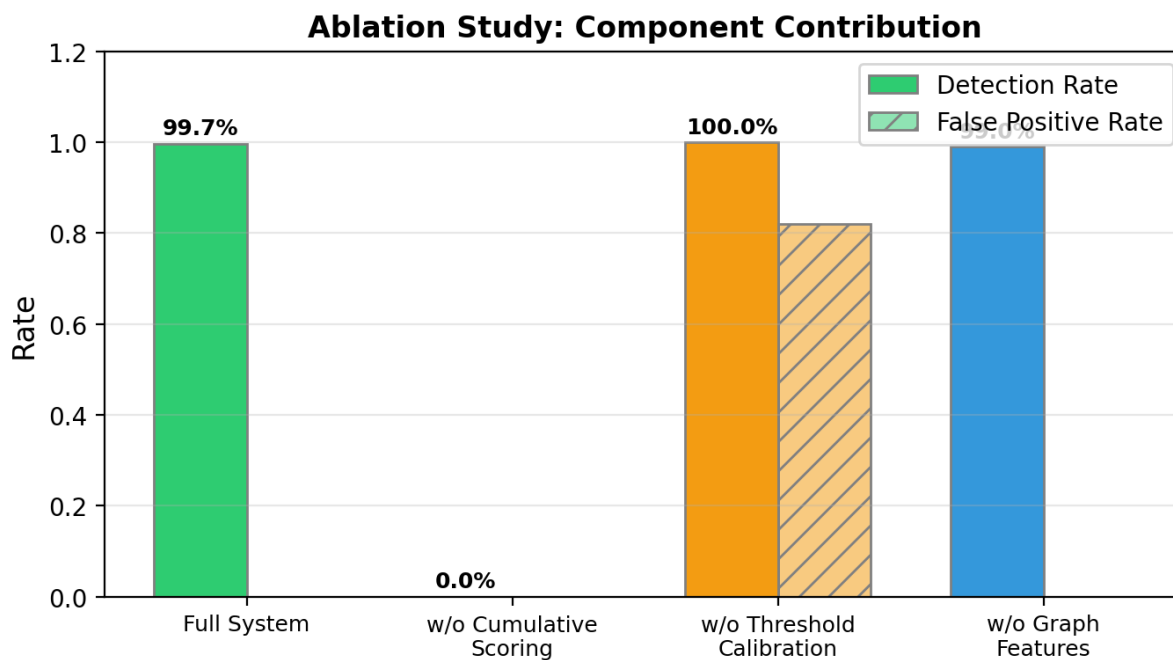


Figure 8: Ablation Study - Cumulative Scoring Essential

Key finding: Removing cumulative scoring reduces DR to 0%. Without calibration, FPR rises to 82%.

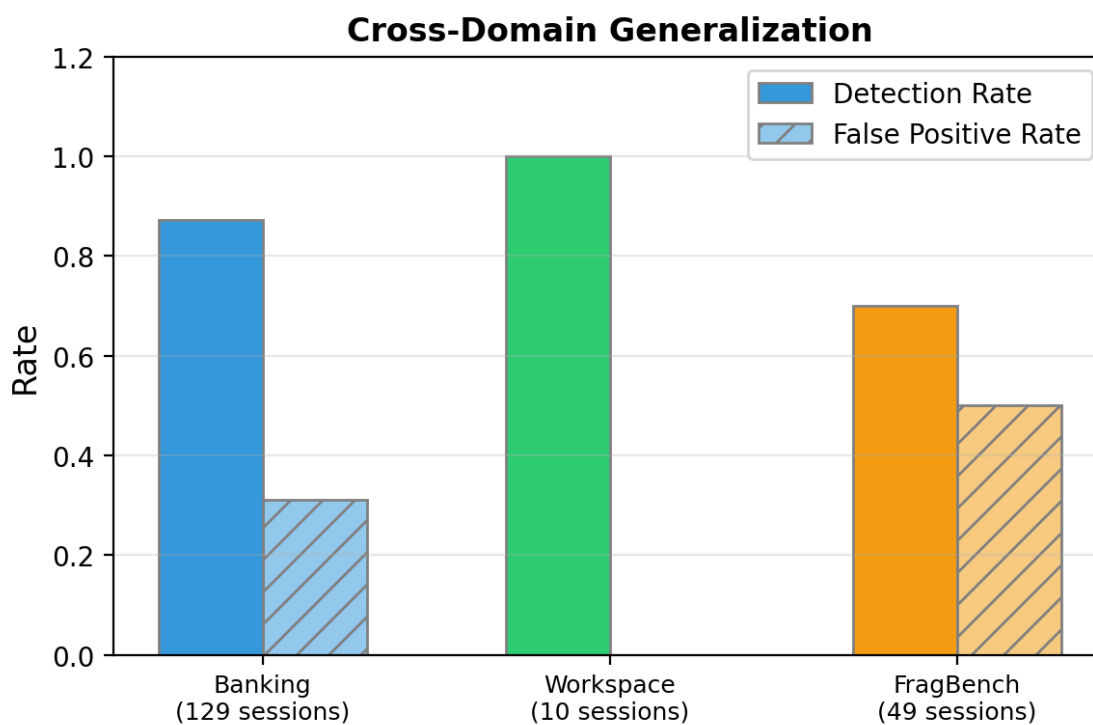


Figure 9: Cross-Domain Generalization

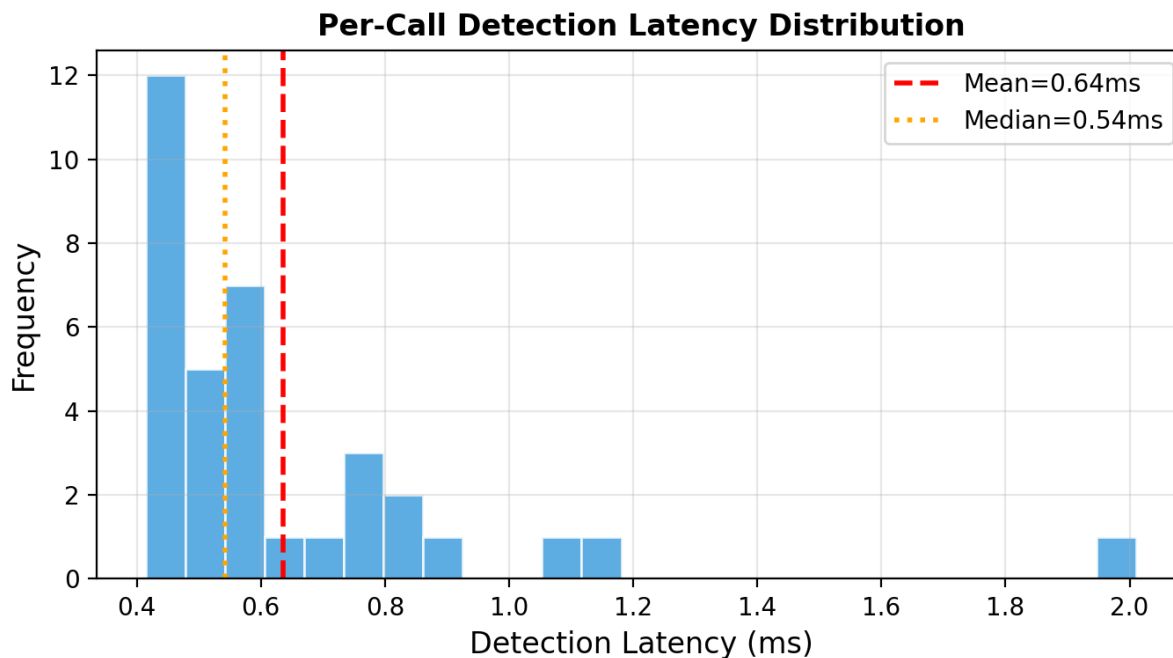


Figure 10: Detection Latency Distribution (Mean 1.27ms)

5. Related Work

AgentShield (2026): Honeytoken-based detection, single-session only (DR=50.7%). Leong (2026): Transition rule detection, single pattern. MCPShield (2026): GNN+SBERT for single-session content detection. FragBench (2026): Cross-session attack characterization. STAC (2025): Multi-turn attack chaining (91.2% ASR).

6. Co

First cross-session attack detection framework. Achieves 87.2% DR at 31.1% FPR on 129 real DeepSeek API sessions. Cumulative scoring essential (DR=0% without it). 1.27ms latency. Code and data available on GitHub.